Paymanshu Sharma

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Experience

Maersk **April 2025 - Present**
Machine Learning Engineer *Bangalore, India*

- Building Shipment CoCaptain: an **agentic AI-powered insights** generator that transforms natural language queries into comprehensive shipment insights, delivering targeted information while automatically enriching user asks to enable deeper business exploration
- Architected a multi-stage **orchestrator-evaluator** pipeline with LangGraph, incorporating planning, decision-making, parallel processing, recursive error handling, LLM-as-a-judge response validation, and citation grounding
- Successful **patent filing**: *Method and System for Optimising Computational Resource Usage during Query Processing* (DKPTO, Apr 2026) for intelligent **context management** in agentic AI workflows: orchestrated, precise state architecting with injectable domain rules, definitions, and formulae for grounding and safety, cutting **per agent/node context by upto 60%** (token count) vs full-state propagation
- Developed and deployed a React frontend with **asynchronous polling** for long-running tasks, **thinking indicators** for progress display, post-response citation tooltips, and **dynamic LLM-generated charts** with Plotly
- Managed significant data volumes from an Azure-hosted PostgreSQL DB. Composite-key **indexing optimizations** for query performance and context-window-aware processing; implementing row- and column-level access control (RBAC) for secure multi-user data access
- Developed an executive dashboard interface web app with AI-generated dynamic widgets and multiple configurable data views over a large database, **Redis caching** and a unified API abstracting multiple data sources; built rapidly applying **spec-driven development** principles while conforming to the Maersk design system

Hike (Rush Gaming Universe) **December 2024 - April 2025**
Machine Learning Engineer *Remote (Bangalore, India)*

- Responsible for developing the AI outreach manager from the ground up - a **real-time voice** based **multilingual** conversational chatbot for automating customer engagement. Integrated a queue-based scheduler to efficiently manage asynchronous call requests at scale via SIP trunking, ensuring scalability and reliability
- Optimized the voice bot's STT-LLM-TTS pipeline using filler responses and interim results to **reduce reply latency**, cutting response time by ∼700ms (30%)
- Set up a configurable template for deploying customized bot agents with modifiable guardrails and speech guidelines. Tested for robustness with varying configs, enabling **personalized user engagement** and offer targeting

Mphasis NEXT Labs (R&D Labs) **December 2021 - December 2024**
Lead Data Scientist (Prev. Data Science Intern) *Bangalore, India*

- Spearheaded development of a structured **information extraction** solution for form documents for enterprise customers. Fine-tuned open-source LLMs with QLoRA, and optimized for **higher inference throughput** with vLLM
- Designed a **multi-agent** conversational architecture with an LLM-powered **recommendation** system employing a two-tower model, zero-shot retrieval, and encoding long-context user and product insights, with **explainable** results
- **Implemented a novel** variant of DeBERTa's Disentangled Attention, incorporating document positional information for a data extraction model trained from scratch. Engineered a task-specific **custom evaluation metric**
- Led the creation and deployment of an intelligent document processing system for **mobile devices**, featuring extraction and RAG-based QA. **Optimized models** using ONNX runtime and quantization, with ObjectBox as the vector database
- Directed the orchestration of a **scalable SaaS product** on AWS cloud services - using CloudFormation, deploying microservices via ECS & Fargate, and automating GPU-enabled EC2 instance scaling with a Lambda-triggered SQS queue
- Delivered applied research spanning a **multi-modal fusion transformer** for audio-video-text retrieval (combinatorial contrastive loss), **distributed federated learning** on edge devices (Flower + AWS Greengrass) for medical image classification, an explainability dashboard with LIME and SHAP-based interpretations, and autoencoder-based **structured data imputation** boosting ∼8% downstream accuracy

MetFlux Research (IIT Bombay) **June 2021 - September 2021**
Computer Vision and Signal Processing Research Intern *Remote (Mumbai, India)*

- Formulated algorithms to estimate a patient's HR, RR, SpO2 and associated **physiological parameters** directly from face & fingertip videos taken by a smartphone camera for a healthcare app under the direction of Prof. K.V Venkatesh
- Optimized **PPG signal quality** by testing signal processing, smoothing and dynamic peak detection algorithms using OpenCV, dlib, and SciPy's signal functions on manually collected test data, achieving reproducible device-agnostic results
- Implemented an image processing and validation pipeline, with a Signal Quality analysis approach to bring about a **55% improvement** in MAE post classical and ML experiments

Amicus.ai **February 2021 - June 2021**
Founding Team Member and Data Scientist *Remote (Mumbai, India)*

- Accurately implemented BERT Named Entity Recognition (NER) models and rule-based text extraction with Regex on over 70% of **scraped unstructured, varied format, legal text data** - from more than 100,000 case files
- Drew up interactive data visualizations using the obtained data, on a web dashboard based on Dash and Plotly

Awarathon **July 2019 - December 2019**

AI Intern *Mumbai, India*

- Devised algorithms utilizing computer vision models for assessing interviewee videos: gaze estimation, gesture recognition, multilingual transcription. Tested on 200+ client clips and incorporated in the primary codebase

Projects and Research Work

Multilingual AAC Communication App | *React Native, Expo, ElevenLabs, Cartesia*

Building an assistive communication (AAC) app with native **Hindi–English code-switching** featuring personalized **next-token phrase prediction** and a **three-tier latency architecture** for TTS (pre-rendered cache, on-demand cached, streaming) balancing instant response against synthesis cost

CereBro: Auditable Opinion Research Engine | *Next.js, React, Supabase, Vercel*

Built a **deep-research** opinion mining engine aggregating sentiments across Reddit, HN, X, and the web where the LLM extracts and clusters claims, and deterministic weighting with full snippet-level source traceability enables instant **zero-LLM-call re-scoring** on source re-weighting

COVID Mythbuster Bot | *Python, TensorFlow/Keras, Flask, nltk, Twilio*

Built a bot using Flask to detect fake news related to COVID-19, conducting ML evaluated with F1 score on a manually collated dataset. Deployed the bot on 3 platforms - WhatsApp, Telegram and as a live web app

Paragon: Crime Detection Center | *Python, TensorFlow/Keras, PyTorch, Streamlit, Pandas, NumPy*

Developed a crime detection center prototype with Streamlit - implementing Face Verification using Haar Cascades and Siamese Networks, Action Recognition from surveillance videos with CNN-LSTM, and statistical region-based crime prediction

iKitchen Aid | *Python, Flutter, TensorFlow/Keras, TensorFlow Lite*

Developed a Flutter app for object detection with YOLOv4 to identify kitchen items for aiding interaction of the visually impaired. Scraped, cleaned and tagged Google Images to create a training dataset of condiment images

AudioAnalytics Package: Emotion Detection | *Python, OpenCV, TensorFlow, Librosa, PyPi*

Published a PyPi Python package to extract audio sample features and predict emotion backed by a 1D CNN model

Gaze Estimation and Saliency Prediction Pipeline to Study Visual Attention for E-Commerce Websites *(Undergraduate Thesis)*

Achievements

LLM Prompt Recovery Challenge: Bronze Medal **[Kaggle Competition]**

Placed in the top 9% of 2200 participants as a solo participant of the LLM Prompt Recovery Challenge, dealing with prompt jailbreaking and evaluation

Bajaj Finserv Hackathon: National-level Runners-Up **[Search and Recommendation Engine]**

Placed 2nd Overall and 1st in category out of 3000+ teams from 200+ colleges nationwide. Developed a Recommendation Engine using BERT embeddings with data mining from the Finserv website to improve their search engine

ACM Mumbai Hackathon: Region-level Winners **[Ideation, prototyping and app development]**

Developed Entrenet – a Flutter app to provide a network for Entrepreneurs, Investors and Job Seekers to connect

SIH Internal Hackathon: College-level Winners **[ML app prototyping and pitching]**

Presented prototypes of an ML-enabled app for aiding farmers through crop detection; Placed 1st at College Level

Technical Skills

Languages: Python, SQL, C++, JavaScript/TypeScript, Dart (Flutter)
ML/AI: PyTorch, Transformers, LangGraph/LangChain, Agentic Systems, RAG, Fine-tuning (LoRA/QLoRA), vLLM, ONNX, Harness Engineering, Evals, Image & Signal Processing, NLP, Multimodal Learning, Recommendation Systems, ETL
Technologies/Platforms: Docker, Microservices, CI/CD, Redis, Git, FastAPI, AWS, GCP, Azure, Linux, Figma

Education

MPSTME, Narsee Monjee Institute of Management Studies, Mumbai **2018 - 2022**

Bachelor of Technology in Data Science – CGPA: 3.82/4.0 *Mumbai, India*

- *Coursework: Data Structures & Algorithms, DBMS, Statistics, Predictive Modelling, NLP, Computer Vision, Applied Math*

Delhi Public School, Gurgaon **2018**

President, NODE (Computer Club) *Gurgaon (NCR), India*

- *Managed a 50+ member club across grades, organized district-level inter-school events*

Misc. Activities

Rotaract Club of Bombay Airport **July 2019 - July 2021**

Initiated and led community development projects, including the Open School program - reaching 150+ underprivileged students, receiving national media recognition. Involved in mentoring for the Technology for Change program

Enactus **July 2019 - July 2020**

On-field research and business venture conceptualization for community upliftment through social entrepreneurship ventures